$L_1 := 1.2\ m$ $L_2 := 0.8\ m$ $L_3 := 1.5\ m$ $L_4 := 1\ m$ $L_5 := 0.5\ m$

$L_6 := 1\ m$ $L_7 := 1.2\ m$ $L_8 := 0.8\ m$ $L_9 := 1\ m$ $L_{10} := 1\ m$

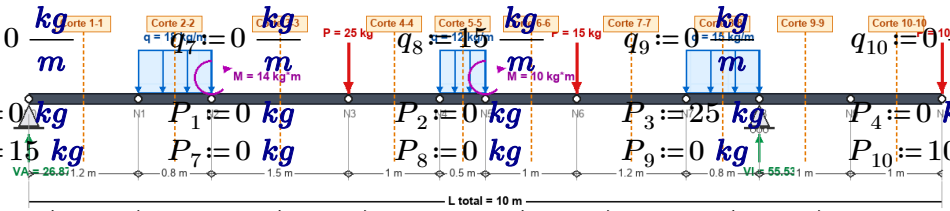
$x_0 := 0.0\ m$ $x_1 := L_1$ $x_2 := x_1 + L_2$ $x_3 := x_2 + L_3$ $x_4 := x_3 + L_4$

$x_5 := x_4 + L_5$ $x_6 := x_5 + L_6$ $x_7 := x_6 + L_7$ $x_8 := x_7 + L_8$ $x_9 := x_8 + L_9$ $x_{10} := x_9 + L_{10}$

$L_{tot} := x_{10}$

$q_1 := 0\ \frac{kg}{m}$ $q_2 := 18\ \frac{kg}{m}$ $q_3 := 0\ \frac{kg}{m}$ $q_4 := 0\ \frac{kg}{m}$ $q_5 := 12\ \frac{kg}{m}$

$q_6 := 0\ \frac{kg}{m}$ $q_7 := 0\ \frac{kg}{m}$ $q_8 := 15\ \frac{kg}{m}$ $q_9 := 0\ \frac{kg}{m}$ $q_{10} := 0\ \frac{kg}{m}$



$P_0 := 0\ kg$ $P_1 := 0\ kg$ $P_2 := 0\ kg$ $P_3 := 25\ kg$ $P_4 := 0\ kg$ $P_5 := 0\ kg$

$P_6 := 15\ kg$ $P_7 := 0\ kg$ $P_8 := 0\ kg$ $P_9 := 0\ kg$ $P_{10} := 10\ kg$

$M_0 := 0\ (kg \cdot m)$ $M_1 := 0\ (kg \cdot m)$ $M_2 := 14\ (kg \cdot m)$ $M_3 := 0\ (kg \cdot m)$ $M_4 := 0\ (kg \cdot m)$ $M_5 := 10\ (kg \cdot m)$

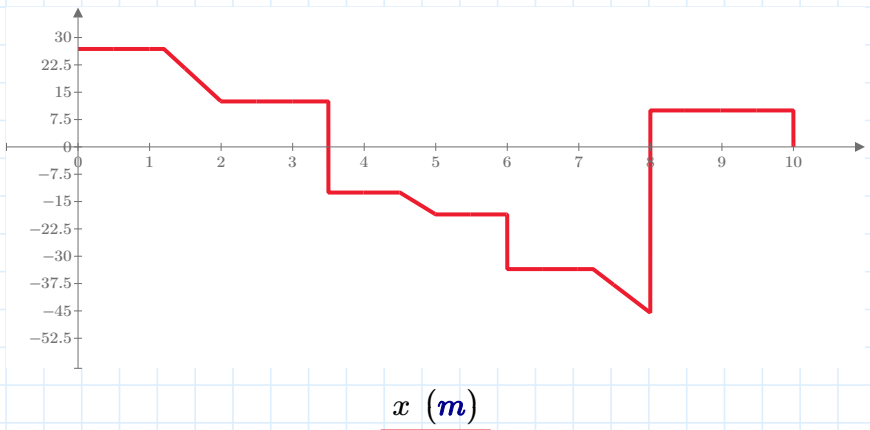
$M_6 := 0\ (kg \cdot m)$ $M_7 := 0\ (kg \cdot m)$ $M_8 := 0\ (kg \cdot m)$ $M_9 := 0\ (kg \cdot m)$ $M_{10} := 0\ (kg \cdot m)$

$V_D := 55.5300\ kg$ $V_A := 26.8700\ kg$

$$M(x) := V_A \cdot x + (x \geq x_8) \cdot V_D \cdot (x - x_8) - (x_2 > x > x_1) \cdot q_2 \cdot \frac{(x - x_1)^2}{2} - (x \geq x_2) \cdot q_2 \cdot L_2 \cdot \left(x - \left(x_1 + \frac{L_2}{2}\right)\right) - (x_5 > x > x_4) \cdot q_5 \cdot \frac{(x - x_4)^2}{2} - (x \geq x_5) \cdot q_5 \cdot L_5 \cdot \left(x - \left(\right.\right.$$

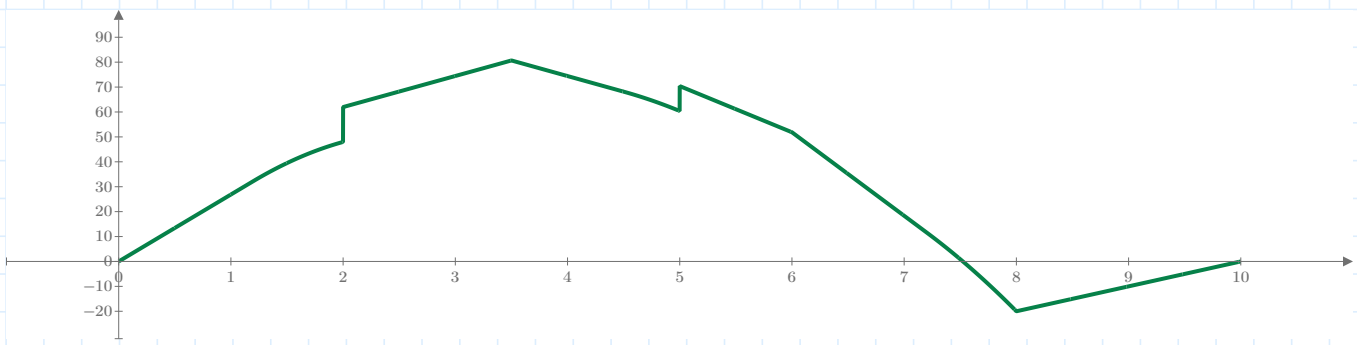
$$Q(x) := V_A + (x \geq x_8) \cdot V_D - (x_2 > x > x_1) \cdot q_2 \cdot (x - x_1) - (x \geq x_2) \cdot q_2 \cdot L_2 - (x_5 > x > x_4) \cdot q_5 \cdot (x - x_4) - (x \geq x_5) \cdot q_5 \cdot L_5 - (x_8 > x > x_7) \cdot q_8 \cdot (x - x_7) - (x \geq x_8) \cdot q$$

$x := 0.0\ m, 0.001\ m .. L_{tot}$



$Q(x)\ (kg)$

$x\ (m)$



$M(x)\ (kg \cdot m)$

$x\ (m)$